

"It's Really Enjoyable to See Me Solve the Problem like a Hero": GenAI-enhanced Data Comics as a Learning Analytics Tool

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ABSTRACT

Data comics are an emergent storytelling format that can enable non-experts to consume salient insights from data. Despite some research investigating the use of comic strips in education, there is limited evidence relating to how students perceive data comics about their own data as a way to reflect on their learning experience. In this paper, we summarise nursing students' perceptions of the advantages and limitations of data comics by presenting personalised Multimodal Learning Analytics (LA) data in data comics form. We present GenAI-enhanced data comic prototypes created using a combination of the generative artificial intelligence tool, Midjourney, and graphics illustration software. Our findings indicate that while students see the potential of data comics as an engaging and enjoyable visualisation technique, concerns remain regarding the perceived lack of professionalism associated with this format.

CCS CONCEPTS

• **Human-centered computing** → **Visualization design and evaluation methods.**

KEYWORDS

learning analytics, data comics, information visualisation

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1 INTRODUCTION & RELATED WORK

Learning analytics (LA) is an emerging area of research focused on collecting and analysing data relating to educational experiences to support educators and students in their reflections and improve their practices [12]. As students often lack the necessary expertise to interpret LA and data, and extract meaningful insights from them [7, 14], LA dashboard interfaces [6] and visualisation techniques – such as data storytelling [3, 7, 13] – have become options to communicate insights and facilitate reflective practices. In their exploration into the different ways *data stories* could be told using visualisations, Segel and Heer [19] proposed seven "genres" to communicate the narrative. One of these is the *comic strip* genre which was identified as a format through which imagery and words could be combined to communicate a compelling narrative [16, 19].

Inspired by the traditional comic strip format [19], *data comics* have emerged as an engaging storytelling format that leverages a combination of panel layouts and narrative content (both illustrative and textual) to convey insights from data [2, 21]. Characterised by the presence of a data visualisation or data-driven narrative, data comics rely on the same judicious choice of *moment*, *frame*, *image*, *word*, and *flow* [16] as traditional comics, while also needing to ensure that the intended insight from the data is clear to both expert and non-expert readers [4, 24]. Bach et al. [4] developed a set of design patterns to aid in the creation and storyboarding of data comics. These design patterns – including narrative, temporal, faceting, visual encoding, granular, and spatial patterns – intend to provide suggested combinations of panel layouts and content such that specific narratives can be told.

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In educational contexts, comics have generally been used as an engagement tool to make potentially complicated learning content more appealing to students [5, 15], including data comics. For example, these have been evaluated in relation to their ability to support student understanding of data visualisation concepts [5, 23]. However, there is limited research on how students perceive data comics depicting their own learning data and analytics. A key challenge is that data comics are commonly crafted manually by a human artist [1, 5]. The literature on data comics has outlined several tools that facilitate this including web applications [9, 26] and custom tools [11, 22]. This makes it challenging to craft personalised data comics that effectively communicate the data of particular students. Yet, recent work by Huang [10] suggested the potential for generative artificial intelligence (GenAI) tools, such as Midjourney, towards at least partly automating the creation of comic strips. This idea could potentially be extended to the creation of data comics as well.

We contribute to this vision by presenting a qualitative study where students were asked to reflect, and provide commentary, on their own educational data presented in various Gen-AI enhanced data comics. The educational data depicted in the data comics was collected in the context of a multimodal LA project where multiple sensing and analytics and AI techniques were used to automatically capture and identify salient aspects of collaboration while students worked in teams in a simulated nursing scenario. The data captured in this simulation (inclusive of positional, voice, and human annotations data) was synthesised into several data comics prototypes by employing the design patterns described by Bach et al. [4]. These prototypes were crafted by using a combination of the GenAI, Midjourney, along with graphic illustrative software. The study contributes both to HCI, Information Visualisation and LA literature by exploring the utilisation of Gen-AI enhanced data comics within a LA space through the following research question: *What are students' perceived advantages and disadvantages of data comics compared to conventional visualisations?*

2 STUDY

2.1 Learning Context

This study focuses on data and participants involved in teamwork simulations conducted in August 2023 as part of an undergraduate nursing degree. The simulations, co-designed by teachers [8], required a team of four nursing students to collaborate and communicate effectively in managing the care of four patients through the following phases:

- **Initial Handover:** Two students were given a briefing on the four patients by their teacher.
- **Initial Assessment:** The students allocated tasks and created a patient care plan together. While enacting this plan the condition of one of the patients would deteriorate, prompting the nurses to escalate the situation by performing a Medical Emergency Team (MET) call - which calls the other two nursing students into the room.
- **Resolving the Emergency:** The student team works together to treat the deteriorating patient (Primary Tasks) while also tending to the other patients (Secondary Tasks).
- **Emergent Diagnosis:** The teacher and the students finalise the simulation by reviewing what happened to determine the cause of the patient's deterioration.

Each simulation lasted around 20.53 min (std = 5.31 min). In each simulation, we collected multimodal data, including audio recordings (captured by Shure PGA31) and positioning data (captured by Pozyx Creator Toolkit). The positioning data was processed to extract the prioritised behaviours (i.e., the percentage of time the nurses spent engaging in working on tasks individually or with their teammates, see the Prioritisation Bar in Figure 1). Positioning data was also combined with audio recordings to show: 1) where the nurses were situated in the simulation room and how much they spoke in that position (see Ward Map in Figure 1), and 2) who each person in the simulation communicated with and the percentage of time they spoke to one another (see Communication Network in Figure 1). Lastly, the audio recordings were transcribed using the OpenAI Whisper medium model [18] and then automatically coded into four critical communication behaviours, (i.e., *acknowledging*, *task allocation*, *escalation*, and *questioning*), using a BERT-based classifier [25]. These coded transcriptions were visualised to show the counts of these behaviours that co-occurred in students' conversations (see Communication Behaviour graph in Figure 1). The teachers debriefed the students using a LA dashboard containing conventional data visualisations depicting the aforementioned data (in Figure 1). Further screenshots of the LA dashboard can be viewed here.

2.2 Participants and Procedure

Students who participated in the above simulation were invited to participate in a follow-up interview to explore the potential advantages and disadvantages of crafting data comics using their multimodal data. A total of 10 participants (P1-10) aged between 20 and 29 (std = 2.67, all female) were interviewed. An hour long Zoom interview was conducted with each of the participants using a "think-aloud" protocol that included the following tasks.

Task 1: Participants were shown their simulation data presented in the teacher's LA dashboard and asked to describe their interpretation of the visualisation, as well as reflect on their team's performance based on the visualisation.

Task 2: Participants were shown seven prototypes of data comics (details below) constructed from the LA dashboard data in Task 1 and following several data comics design patterns [4]. Each participant viewed the prototypes in the same order. They were asked to describe what they understood about the data comics and whether or not the data comics helped them reflect on their performance. After being exposed to all of the prototypes, the participants reflected on the advantages and disadvantages of data comics compared to the conventional dashboard visualisations they were exposed to in Task 1.

3 DESIGN OF THE DATA COMICS PROTOTYPES

3.1 Data Comics Design Patterns

Data comics design patterns are defined by Bach et al. [4] as "a set of panels with specific layout and content relation" with the intention of

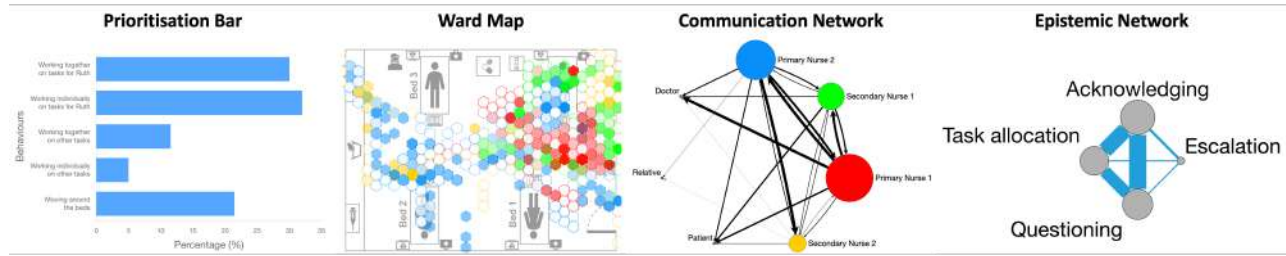


Figure 1: Visualisations from the original learning analytics dashboard

purposefully leveraging narrative elements in order to convey the data story. They comprise two key dimensions (i) the *panel layout* – the framing of panels that determines how readers are guided through the comic – and (ii) the *content relation* which describes the placement of, and relationship between, narrative elements within the panels in order to tell a meaningful and cohesive story [4]. The following subsections summarise how the design patterns described by Bach et al. [4] were utilised to formulate the data comics prototypes shown in Figures 2, 3, and 4.

The data comics prototypes were created in Clip Studio Paint Pro (v1.12.8) using images generated in Midjourney (Niji Model v5). In addition to the hospital ward backgrounds, the Midjourney GenAI tool was used to create nurse and doctor characters with various poses and emotions to embrace the expressive nature of the comic medium while ensuring consistency between panels. Using the Niji Model enabled rapid generation of a range of characters in a similar art style. Due to current Midjourney limitations – particularly its challenges in creating a cohesive and coherent comic strip with text elements – the AI-generated characters and backgrounds were arranged into panels using Clip Studio.

3.2 Learning Scenario Prototype

Learning Scenario depicts the simulation that students participated in as part of their nursing studies. The most stylised of the data comics prototypes presented in this study, *Learning Scenario* focuses on giving students an overview of the events that transpired in lieu of presenting specific statistics relating to the student or their team. This prototype most closely resembles the *Exposé* design pattern proposed by Bach et al. [4] as the emphasis is on providing information through narrative elements rather than data. Additional design patterns within this prototype can be viewed here.

3.3 Prioritisation Prototypes

The *Prioritisation* prototypes aimed to illustrate the percentage of time the nursing students spent engaging in different behaviours (e.g., working individually on a specific patient, working together on other tasks, etc.). Two visually distinct prototypes were created to depict this data:

Prioritisation 1: This prototype integrated a bar chart accompanied by a data comic detailing the percentage of time in an annotation along with a visual representation of the behaviour (Figure 3-left). The main design patterns in this prototype is *Overview & Detail* [4], as individual bars in the bar chart in the lower half of the data comic are explained or depicted in greater detail by the data story within the speech bubble. *Multiple facets* [4] is also relevant to

this prototype as the top half is using textual elements and images to represent the same data set shown in the bar chart.

Prioritisation 2: Adapts the bar chart visualisation from the LA dashboard into a comic by visually encoding the size of the panels to the size of the bars (Figure 3-right). This was crafted using the *Percentage Panel* pattern [4].

3.4 Proximity Prototypes

The *Proximity* prototypes integrate the “ward map” data visualisation that shows where the nursing students were over the course of the simulation, as well as a pictorial representation of the behaviours that the nurses were engaging in at that particular point in time. There were two prototypes:

Proximity 1: Mainly comprising the conventional visualisation of the ward map from the LA dashboard, data comics elements contained within speech bubbles illustrate what is happening at each hexagon cluster (Figure 4-left). This prototype also uses the *Overview & Detail* pattern while also incorporating the *Legend* visual encoding pattern [4].

Proximity 2: The comic is divided into three large panels with the middle showing the ward map from the LA dashboard, and the other two showing two important tasks that are occurring in parallel (Figure 4-right). This layout more closely resembles a traditional comic strip; yet, the consistent background in the top and bottom panel is similar to the polyptych pattern (where similarity between panels is used to suggest that they take place in the same world) [4].

3.5 Communication Network Prototype

Wang et al. [22] believed that incorporating animations into data comics could allow readers to identify change. This *Communication Network* comic utilises animation to show the changes in the communication network through each stage of the simulation (Figure 5-left).

3.6 Epistemic Network Prototype

The *Epistemic Network* prototype uses the *branching* layout described by Bach et al. [4] to extend on the epistemic network diagram from the LA dashboard. It utilises the *Overview & Detail* design pattern [4] as each panel illustrates the corresponding node of the epistemic network diagram with a textual example of *Questioning*, *Acknowledging*, *Task Allocation*, and *Escalation* (Figure 5-right).

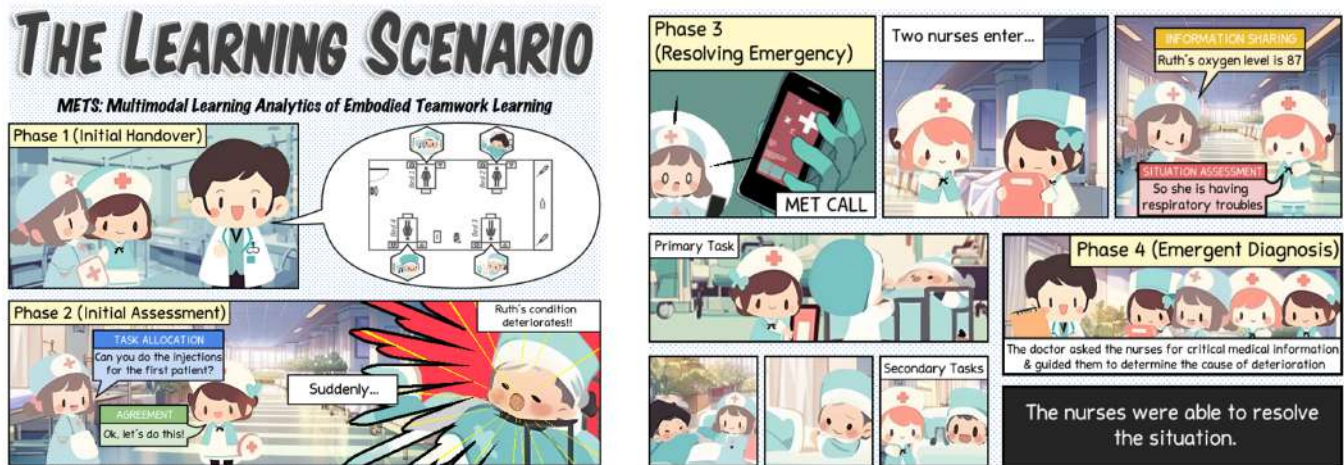


Figure 2: The Learning Scenario data comic prototype

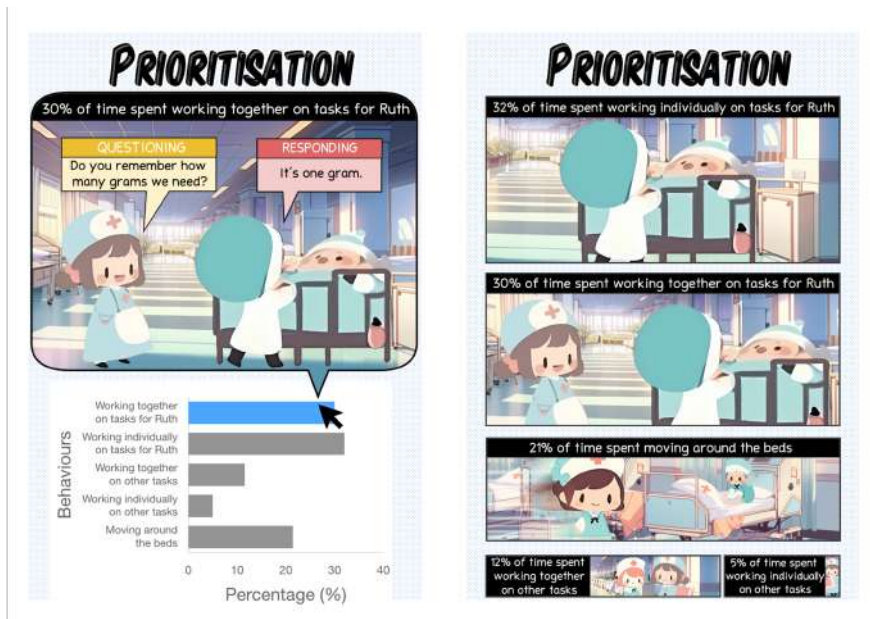


Figure 3: The Prioritisation data comic prototypes

4 PRELIMINARY ANALYSIS AND RESULTS

4.1 Preliminary Analysis

The interviews were recorded and fully transcribed electronically using the One AI Language Studio. An inductive thematic analysis was conducted and discussed by two researchers resulting in two emergent themes of perceived *advantages* and *disadvantages* of GenAI-enhanced data comics as LA tools.

4.2 Advantages of Data Comics

The primary benefit that participants discussed was the increased motivation to look at the visualisation and reflect on the message it was conveying and/or their own performance. The majority of participants praised the data comics as *“fun”* with one participant

describing the comics as *“more ... enjoyable [and] interesting”* (P2) than conventional visualisations. P2 elaborated on this by noting *“I’m pretty happy seeing me as the main character ... It’s really enjoyable for me to see me ... inside this comic and solving the problem like a hero”*.

In comparison to conventional visualisations, participants noted that data comics *“capture the audience a lot better”* (P4) as they are *“more engaging ... [and] ... more accessible”* (P5) than the *“daunting”* (P7) visualisations they were exposed to in Task 1. Many of the nursing students thought data comics were appropriate for *“visual learner[s]”* (P1, P4, P10) as they *“make the information ... easier to digest rather than straight data”* (P8).

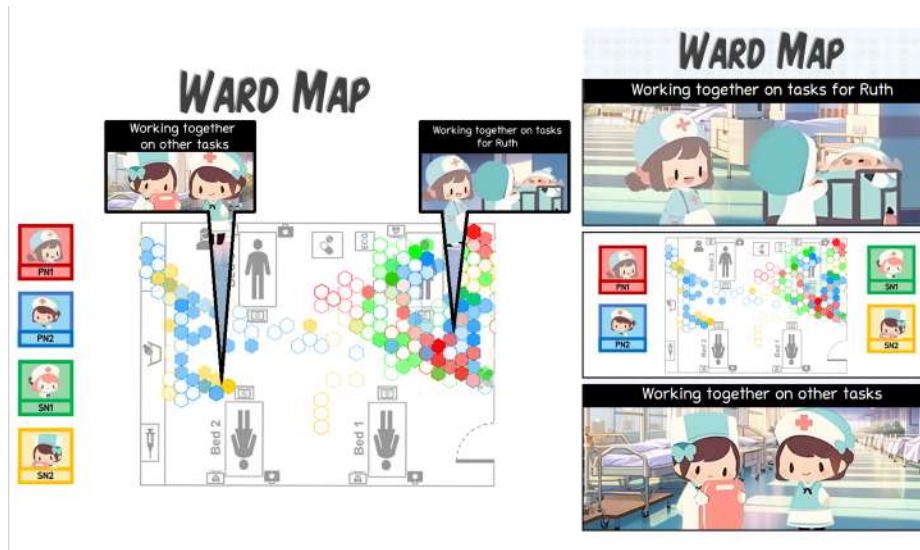


Figure 4: Proximity Prototypes

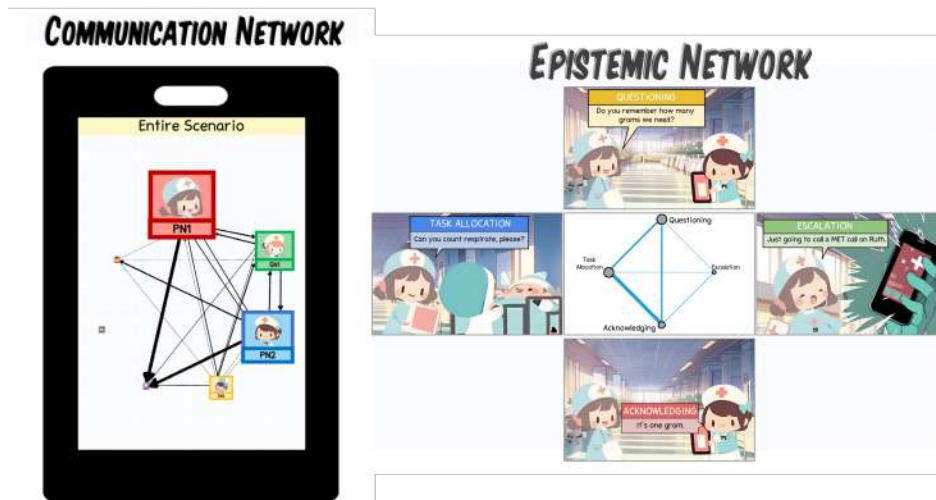


Figure 5: The Communication Network [left] & Epistemic Network [right] data comic prototypes

4.3 Disadvantages of Data Comics

The main criticism of data comics was that they lack professionalism in the particular context of higher nursing education. P1 mused that *"I see nursing as ... professional ... you're trying to save someone's life. And then I think of cartoons [as] ... immature"*. There was a belief that data comics *"take away the intensity of the situation"* (P8) and that *"health industries and universities wouldn't like it"* (P5) being used as a source.

The artistic nature of the data comics prompted some participants to make commentary on their designs. Participants warned that thoughtless applications of the data comics ideas and elements may lead to *"extra noise"* (P3) that may cause *"key information [to] be missed or less obvious"* (P6). When commenting on the design of the characters within the data comics, P3 was critical of the nurse characters reinforcing antiquated gender roles stating *"it just shows*

that the male should be the doctor and then the girl should be the nurse".

5 DISCUSSION

5.1 Potential Value of Data Comics

In their reflections on the benefits of data comics, participants expressed that engagement and *"fun"* factor were the main attraction of this format. This finding is consistent with other explorations into data comics by Wang et al. [24] and research into applications of traditional comic strips in an educational setting [15]. It is thought that this appeal of data comics may be due to people being familiar with comics [5]. This idea of familiarity was discussed in a data visualisation context by Knaflitz [13] who suggested that non-data experts may be reluctant to analyse conventional visualisations, that they may be unfamiliar with, as they may perceive them as

too complicated. As highlighted by Paulsen and Lindsay [17], LA research has investigated the utility of data visualisations and LA dashboards to support students. However, there is still uncertainty surrounding how these can be employed so that students can best understand their own learning data and use the insights to improve their learning strategies [17].

Matuk et al. [15] suggested that a primary benefit of comics in learning was that they can enable students to relate to the depicted characters and situations on an emotional level. In our study, while most of the participants commented broadly on the data comics being able to capture their attention better than conventional visualisations, one student was able to relate to the data comic content on such a level that they projected themselves as the “*main character*”. This presents an opportunity for future data comics research to assess whether students can engage to a greater level with the insights if the physical characteristics of characters resemble the students themselves.

5.2 Limitations of Data Comics

A common view amongst participants was that data comics lacked the necessarily professionalism to be applied in tertiary nursing education. The resistance to the adoption of comics in educational settings has also been shared by teachers in Matuk et al. [15]. It is possible that the “*cute*” design (versus a more realistic style) of the characters in our data comics may have contributed to perceptions of unprofessionalism or informality. Future investigations will need to be undertaken to assess the viability of data comics to support students and educators in other contexts or with alternative visual designs. Moreover, conventional data visualisations, as well as data storytelling enhanced visualisations, often rely on the use of pre-attentive attributes to capture and guide the attention of readers [13], suggesting visualisations need to be decluttered. In our study, the participants’ expressed concern that data comics may introduce unnecessary visual “*noise*” that may serve to obscure the main insight. This suggests that – although data comics may have a more stylised design than conventional visualisations – designers should take care to avoid excessive extraneous visual elements to allow readers to focus only on the important information [13].

5.3 Data Comics Design Recommendations

Based on the comments from the nursing students, the following preliminary design recommendations can be suggested for informing future research:

- The designed data comics should balance the visual elements (i.e., colours, panels, characters, etc.) such that the main takeaway is clear while also preserving the enjoyability and engagement factor of the format.
- The characters and scenarios within the comic should be contextually appropriate to avoid dissonance between the content of the comic and the context of the data from which it was derived.

Additionally, a criticism of the data comics prototypes was the prevalence of female characters. It is a known limitation of image generation models that they tend to reinforce stereotypes (e.g., images of nurses are biased towards females) [20]. Therefore designers who utilise GenAI tools in the creation of data comics should be

cautious to avoid biases towards particular genders, skin tones, or otherwise in their characters.

5.4 Limitations of the Study

A limitation of this study is the small sample size of ten participants, along with all participants being female. This means that the results cannot be generalized to the wider population. To best assess the potential of data comics in aiding nursing students, the sample should be increased and include male students. To determine the suitability of data comics for LA, future works should also aim to include a variety of students from different disciplines and backgrounds.

6 CONCLUDING REMARKS

In this paper, nursing students were exposed to data comics prototypes that depicted their LA data in order to investigate the advantages and disadvantages of this format. The results indicated that the reception to data comics was mixed. Participants praised data comics as an engaging medium that was more “*fun*” compared to conventional visualisations. However, criticism was directed at the perceived lack of professionalism associated with comics used in tertiary nursing education, as well as the potential for the data comic features to add unnecessary noise and obscure salient insights.

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